Fernando Daniel García

Over forty years building software, in science and in business.
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Profile

Computer scientist with a career in two worlds: scientific research, at CONICET (Argentina's National Scientific and Technical Research Council) since 1984, where he has developed software for mathematics, ecology, energy and the atmospheric sciences; and business, where he founded technology companies for different markets. Three traits recur throughout: an interest in new technologies as soon as they appear, building his own tools, and carrying ideas from one field to another.

Innovation

- **XG Tecnología (1993–1995):** interactive presentations and kiosks with games for Ford, Volkswagen, Toyota, Sancor and Telecom, among others, built on his own engine on top of the hardware manufacturers' libraries rather than the "multimedia" tools of the time.
- **García & Strauss (1995–2004):** games and interaction built into business applications, such as Car One's self-service information terminals.
- **FT WAVE (2000–2002):** design of an online system for distributing new music releases to radio stations across Argentina over dial-up connections, with its own digital music player, hired by Sony Music, EMI Music, Warner Music, BMG, Universal Music and Leader Music.
- **Digidramamancia (2005):** software that generated the theatre script and the musical score in real time from the audience's fingerprints, for a piece premiered at Tecno Escena 05 (Centro Cultural Borges, Buenos Aires).
- **Agilmind (2009):** *Axe Music Star*, an interactive game for Axe that ran on a desktop computer's webcam: adaptive music, computer vision, genetic algorithms and real-time non-linear video.
- **Agilmind today:** artificial intelligence (AgilPost, AgilBank, AgilFace), electronic signature (AgilSign), credit, tender and document management (AgilCred, AgilLic, AgilDoc) and Jupiter, a platform for the music industry and copyright.

Own tools

The languages Augur (expert systems), ShowMe (multimedia presentations) and Haiku (full-stack web applications); his own inference engine for expert systems (1989) and an LR/LALR parser generator (1990).

Science across fields

- **Mathematics:** symbolic artificial intelligence (symbolic differentiation in Prolog) and network reliability.
- **Ecology:** population dynamics models and expert systems for natural resource management, with his own inference engine.
- **Nuclear energy (CNEA, 1987):** a rule-based prototype to detect abnormal events in the RA3 reactor.
- **Atmospheric sciences (CIMA-IFAECI):** satellite (GOES-16) and radar data, bots that collect data from automatic weather stations, community-based flood early warning, and a severe weather event registry for five countries (JICA–SMN).

Recognition

A GECCO 2000 paper cited as prior art in 16 US patents; co-author in the *Bulletin of the American Meteorological Society* (2024); dataset published at NCAR (2019).

Career

- **CONICET, since 1984:** Instituto Argentino de Matemática, FCEN-UBA, CEPAVE (CONICET-UNLP) and CIMA-IFAECI (CONICET-UBA).
- **Companies:** co-founder of XG Tecnología (1993–1995), García & Strauss (1995–2004), FT WAVE (2000–2002) and Agilmind SRL (since 2006); consultant at Borland Argentina (2001–2006).

Music

He has sung in folk groups since the age of 12 (among them Inti Huasi, which he founded), was a member of the Grupo de Experimentación Vocal (GEV), wrote arrangements for the Fundación San Telmo choir and has composed music for theatre, such as *Vanya en miniatura* (2025). Solo album: *Ronda* (2015). Registered as author, performer and phonogram producer with SADAIC, AADI and CAPIF. Music at https://fidogarcia.bandcamp.com.

Education

Programador Superior (computer programming degree), Universidad Nacional de San Luis (1983). *Genetic Algorithms in Search, Optimization, and Machine Learning*, University of Illinois at Urbana-Champaign, with David E. Goldberg (2000).